DOD-I-51323C
11 February 1985
SUPERSEDING
DOD-I-51323B(EA)
17 April 1979

MILITARY SPECIFICATION

INSECTICIDE, DICHLORVOS, IMPREGNATED STRIP (METRIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of dichlorvos insecticide strip. Dichlorvos is chemically known as 2,2-dichlorovinyl dimethyl phosphate.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	-	Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
PPP-B-585	-	Boxes, Wood, Wirebound
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-F-320	-	Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament, Munitions and Chemical Command, ATTN: AMSMC-TDC-S (A), Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6840

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MILITARY

MIL-P-116 - Preservation, Methods of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

CODE OF FEDERAL REGULATIONS (CFR)

40 CFR 162 - Regulations for the Enforcement of the Federal Insecticide, Fungicide, and Rodenticide Act
49 CFR 171 to 179 - Department of Transportation Hazardous Materials Regulations

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publications should cite "40 CFR 162" or "49 CFR 171 to 179.")

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

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ASTM STANDARDS

E 203 - Water Using Karl Fischer Reagent (DoD Adopted)

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Composition. Dichlorvos strips shall be prepared from technical grade dichlorvos insecticide conforming to the characteristics of table I, which has been combined with polyvinyl chloride and suitable plasticizers to form a homogeneous, firm, solid mass. The contractor shall certify that this requirement has been met.

TABLE I. Characteristics of technical grade dichlorvos insecticide used to prepare strips

Characteristic	Minimum	Maximum
2,2-dichlorovinyl dimethyl phosphate, percent by weight	96.0	----
Water content, percent by weight	----	0.01
Acidity, milliequivalents of NaOH per gram	----	0.01

3.2 Physical characteristics. Dichlorvos strips shall conform to the physical characteristics of table II when examined as specified in 4.2.3.2.

TABLE II. Physical characteristics of strips

Characteristic	Minimum	Maximum
Length, millimeters	155	173
Width, millimeters	53	shall fit in holder
Thickness, millimeters	7.0	9.2
Weight, grams	80	115

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3.3 2,2-Dichlorovinyl dimethyl phosphate content. Dichlorvos strips shall contain no less than 18.6 and no more than 22.0 percent by weight 2,2-dichlorovinyl dimethyl phosphate when tested as specified in 4.2.4.1.

3.4 Water content. Dichlorvos strips shall contain no more than 0.2 percent by weight of water when tested as specified in 4.2.4.2.

3.5 Length of effectiveness. Dichlorvos strips shall be effective in use for a period of no less than 120 days. The contractor shall certify that this requirement has been met and shall state how the 120-day effectiveness was determined.

3.6 Holder and hanger. Each dichlorvos strip shall be furnished with a holder and a hanger.

3.7 Labeling. Unit containers shall be labeled in compliance with the requirements of 40 CFR 162 and 49 CFR 171 to 179. Labels shall be approved and registered in accordance with 40 CFR 162. As a minimum, each container shall be labeled with the date of manufacture and the following information:

(a) The name and percent by weight of each active ingredient
(b) Percent by weight of inactive ingredients
(c) National stock number, military specification number, contract number, and lot number
(d) Manufacturer's name and address
(e) EPA registration and establishment number
(f) Directions for use:

To control flies, mosquitoes, gnats and other small flying insects in enclosed areas, use one strip for each 28 cubic meters (1,000 cubic feet) of enclosed area.

To control flies, mosquitoes, gnats and other small flying insects in horse barns and dog kennels, place one strip for each 28 cubic meters (1,000 cubic feet) of space. Do not contaminate food, water, or foodstuffs.

For the control of flies in garbage cans, attach one strip inside can and replace lid.

For the control of mosquitoes breeding in catch basins, suspend one strip approximately 250 millimeters (10 inches) above the water line in each basin.

(g) Precautionary information:

KEEP OUT OF REACH OF CHILDREN!

WARNING! HARMFUL IF SWALLOWED!
CAUSES EYE AND SKIN IRRITATION!

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After prolonged storage, a small amount of liquid may form on the strip. Avoid eye and skin contact with liquid. Wash hands thoroughly with soap and water after handling strip.

CAUTION! Do not place in hospitals or clinic rooms, such as patient rooms, wards, nurseries, operating and emergency areas.

Do not use in any areas where infants, the ill, or the aged are or will be present for an extended period of confinement.

Do not use in kitchens, restaurants, or places where food is prepared or served.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the dichlorvos strips which have been prepared from one batch of technical grade dichlorvos insecticide and one batch of matrix material and which has been produced by one manufacturer, at one plant, under the same manufacturing conditions.

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling for examination of packaging shall be conducted in accordance with MIL-STD-105 using an AQL of 2.5 percent defective.

4.2.2.2 For examination of strips. Sampling for examination of strips shall be conducted in accordance with MIL-STD-105, inspection level S-4 using an AQL of 4.0 percent defective.

4.2.2.3 For test. Five of the strips taken for examination in 4.2.2.2 shall also be subjected to testing in accordance with 4.2.3.3.

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4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one unit, intermediate, or shipping container, as applicable, ready for shipment. Sample unit, intermediate, and shipping containers shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged
(e) Holder and hanger missing or not packed with strip
(f) Fiberboard packing aids missing or not as specified (where required)
(g) Unitization not as specified
(h) Marking or labeling incorrect, missing, or not as specified
(i) Strip not packaged in accordance with method IA-8 of MIL-P-116

4.2.3.2 For examination of strips. The sample unit shall be one strip. The strips taken in 4.2.2.2 shall be examined for the following defects using an AQL of 4.0 percent defective:

(a) Length of strip not as specified
(b) Width of strip not as specified
(c) Thickness of strip not as specified
(d) Weight of strip not as specified

4.2.3.3 For test. The five strips taken in 4.2.2.3, after having been examined in accordance with 4.2.3.2, shall be tested as specified in 4.2.4.1 and 4.2.4.2. Failure of either test by any of the five strips shall be cause for rejection of the lot represented.

4.2.4 Tests. Reagent grade chemicals shall be used throughout the tests. Tests shall be conducted as follows:

4.2.4.1 2,2-Dichlorovinyl dimethyl phosphate content.

4.2.4.1.1 Equipment and apparatus. Determine the percent by weight 2,2-dichlorovinyl dimethyl phosphate in the dichlorvos strips by means of gas-liquid chromatography using a flame ionization detector. Recommended column parameters and chromatographic conditions are shown in table III. To obtain good peak shape, adequate resolution, and appropriate retention times, these conditions may be modified.

4.2.4.1.2 Calibration. Prepare standard solutions of 2,2-dichlorovinyl dimethyl phosphate by weighing to the nearest 0.1 milligram (mg) 12.5, 25, 37.5, 50, and 62.5 mg quantities of standard 2,2-dichlorovinyl dimethyl phosphate of known purity (see 6.4) in 25-milliliter (mL) volumetric flasks. Dilute to the mark with ACS grade chloroform and mix. The standard should be stored in the refrigerator and equilibrated to room temperature before opening.

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TABLE III. Chromatographic conditions

Characteristic	Condition
Column material	Pyrex glass
Column dimensions	1.83 meters (2 mm ID)
Column coating and support	3 percent OV-17 on 100/120 "Chromosorb WHP" or 3 percent OV-120 on 100/120 "Chromosorb" or 1.5 percent OV-17, 1.95 percent QF-1 on 80/100 "Gas-Chrom Q"
Carrier gas	Nitrogen
Carrier gas flow rate	20 to 60 mL/min (207 kilopascals head pressure)
Oven temperature	125°C
Injector temperature	170°C
Detector temperature	250°C

Make duplicate 5-microliter injections of the standard solutions into the gas chromotograph using a 10-microliter syringe. Determine the peak area for each injection and plot area versus milligrams 2,2-dichlorovinyl dimethyl phosphate per milliliter in the standard solution. A line connecting the points should yield a straight line. Recalibrate if the detector response is suspect. Weigh to the nearest 0.1 mg, 0.2 grams (g) 2,2-dichlorovinyl dimethyl phosphate into a 125-mL Erlenmeyer flask. Add 100.0 mL chloroform, stopper, and mix (yields 2 mg per mL or 2 micrograms per microliter solution). Be certain to take 2,2-dichlorovinyl dimethyl phosphate purity of the standard into account when determining 2,2-dichlorovinyl dimethyl phosphate concentrations. Determine the concentration of 2,2-dichlorovinyl dimethyl phosphate in the standard solution as follows:

$$\text{concentration of standard solution in 2,2-dichlorovinyl dimethyl phosphate per milliliter} = \frac{AB}{C}$$

where: A = Milligrams of standard used,
B = Known purity of the standard, and
C = Milliliters of chloroform used to dilute the standard.

4.2.4.1.3 Procedure. Dice the impregnated strip into 100-millimeter or smaller pieces using a belt (leather) cutter, vinyl tile cutter, razor blade or other means. Weigh into a 125-mL Erlenmeyer flask about 500 mg of diced strip to the nearest 0.1 mg. Using a 50 mL pipet, add 50.0 mL of chloroform to the flask and close with a ground glass stopper. Place on a shaker for 4 hours at a speed sufficient to keep the contents agitated but not splashing up onto the stopper. Inject identical volumes (5 microliters) of the standard and sample solutions in the following sequence: 1 standard, 2 sample, 1 standard, etc. Syringe filling technique should include a 1-microliter rinse as follows: Rinse syringe in chloroform, depress plunger all the way, draw in 1 microliter of air, 1 microliter of chloroform, 1 microliter of air, 5 microliters of

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sample or standard and 1 microliter of air. Wipe the needle with a tissue prior to injection. Even though the calibration curve prepared above shows linearity over the range of concentrations used, the standard peak areas should not differ from the sample peak areas by more than 10 percent for the most reliable results.

4.2.4.1.4 Calculations. Determine the peak heights or areas. Use the values obtained in the equation given below. Calculate the percent by weight 2,2-dichlorovinyl dimethyl phosphate in the sample as follows:

$$\text{Percent by weight 2,2-dichlorovinyl dimethyl phosphate} = \frac{ACD}{10\ (BE)}$$

where: A = Peak height or area of 2,2-dichlorovinyl dimethyl phosphate in sample,
B = Peak height or area of 2,2-dichlorovinyl dimethyl phosphate in reference standard,
C = Concentration of 2,2-dichlorovinyl dimethyl phosphate in reference standard solution, in milligrams per milliliter,
D = Volume of sample solution in milliliters (take into consideration aliquot factors, if any), and
E = Weight of sample in grams.

4.2.4.2 Water content. Determine percent by weight water content in accordance with ASTM E 203.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packing. One dichlorvos strip shall be unit packed in accordance with method IA-8 of MIL-P-116. The unit packed strip plus its holder and hanger shall then be packed together in a waterproof bag in accordance with method IC-3 of MIL-P-116 to make one complete unit pack.

5.1.1.2 Intermediate packing. Forty-eight dichlorvos strips unit packed as specified in 5.1.1.1 shall be intermediately packed in a uniform, compact pattern in a close-fitting fiberboard box conforming to grade V3c of PPP-B-636. Fiberboard pads conforming to grade W5c or V3c of PPP-F-320 shall be inserted in the box as needed to prevent motion of the contents. The box shall be closed in accordance with method IV in the appendix to PPP-B-636.

5.1.2 Level C unit packing. One dichlorvos strip with its holder and hanger shall be packed together in accordance with method IA-8 of MIL-P-116 and Department of Transportation (DOT) regulations.

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5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. Two intermediate packs of dichlorvos strips prepared as specified in 5.1.1.2 shall be packed in a close-fitting wood box conforming to, at the option of the contractor, class 2, style 4 of PPP-B-621; type overseas, style optional, grade B of PPP-B-601; or class 3, style optional of PPP-B-585. The box shall be designed for a type 2 load. Fiberboard pads formed from material conforming to grade W5c or V3c of PPP-F-320 shall be inserted in the box as needed to prevent motion of contents. The box shall be closed and reinforced as specified in the appendix to the applicable box specification.

5.2.2 Level B. Two intermediate packs of dichlorvos strips prepared as specified in 5.1.1.2 shall be packed in a close-fitting fiberboard box conforming to style RSC, grade V3c of PPP-B-636. Sheets of fiberboard conforming to grade W5c or V3c of PPP-F-320 shall be placed between the contents of the box and the inside faces of the box, where necessary, to assure a tight pack. The box shall be closed in accordance with method III as specified in the appendix to PPP-B-636.

5.2.3 Level C. Dichlorvos strips, unit packed together with holders and hangers, shall be packed in a manner to assure acceptance by common carrier and safe arrival from source to initial destination. Containers shall be in accordance with DOT regulations and the Uniform Freight Classification Rules and National Motor Freight Classification Rules and the rules and regulations applicable to any other intended mode of transportation.

5.3 Unitization. Level A and B shipping containers shall be unitized in accordance with MIL-STD-147 using load type I and a soft wood pallet conforming to type IV of NN-P-71. Level C shipping containers shall be unitized in uniform quantities in a manner to expedite handling, reduce labor, and to assure protection of contents in transit and stable stacking in storage.

5.4 Marking. Level A and B unit packs, intermediate packs, packing, and unitized loads shall be marked in accordance with MIL-STD-129 and DOT regulations. Level C unit packs, packs, and unitized loads shall be marked in accordance with MIL-STD-129 and DOT regulations and in a manner to assure easy identification of contents and shipper and addressee.

6. NOTES

6.1 Intended use. Dichlorvos strips are intended for use against flies, mosquitoes, gnats, and other small flying insects in enclosed areas to include troop housing, non-food areas of food handling establishments, retrograde cargo, and various animal buildings. In addition, the strips are intended for use against flies in garbage cans and against mosquitoes in catch basins.

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6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Level of preservation required (see 5.1), and
(c) Level of packing required (see 5.2).

6.3 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E 29.

6.4 2,2-Dichlorovinyl dimethyl phosphate standard. 2,2-Dichlorovinyl dimethyl phosphate of known purity for use in preparing a calibration curve in the assay test may be obtained from the Shell Chemical Company, One Shell Plaza HS & E Product Safety and Compliance, P. O. Box 4320, Houston, TX 77201.

Custodian:

Army - EA

Review activities:

Army - MD, GL
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Preparing activity:

Army - EA

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions -- Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
DOD-I-51323C	INSECTICIDE, DICHLORVOS, IMPREGNATED STRIP (METRIC)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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